ContamCal: Post-Hoc Calibration of Benchmark Scores under Training Data Contamination via Contamination-Augmented Item Response Theory

Anonymous

May 13, 2026

Abstract

TODO: Abstract to be written after experiments.

1 Introduction

Large language model (LLM) evaluation increasingly relies on standardized benchmarks to compare models and track progress, yet mounting evidence reveals that these benchmarks are systematically compromised by training data contamination [Chang et al., 2025]. Yang et al. [2026] demonstrate that rephrased benchmark items appearing in training corpora inflate MMLU accuracy by 8–15 percentage points while evading standard n-gram decontamination, and that reinforcement learning (RL) post-training with GRPO further amplifies contamination-driven gains while simultaneously concealing the detection signals that would reveal them. Dose-response analyses confirm a monotonic relationship between contamination exposure and score inflation, with effects persisting across model scales [Dong et al., 2026]. Even minor perturbations to benchmark item surface forms—which should not affect genuinely capable models—produce substantial accuracy drops, questioning the validity of reported scores [Wang et al., 2025]. Together, these findings point to a reliability crisis in LLM evaluation: reported benchmark scores increasingly conflate memorization with generalizable capability [Rao et al., 2026].

Several methods have been developed to detect benchmark contamination. Min-K%++ [Duan et al., 2025] and Time Travel [Oren et al., 2024] perform membership inference against pre-training corpora; ConStat [De Koninck et al., 2024] measures performance gaps between original and paraphrased items; Self-Critique [Wüst et al., 2025] identifies memorization through entropy collapse in model self-evaluations; and KDS [Xu et al., 2025a] uses knowledge distillation signatures to flag leaked items. While these methods can detect contamination, none produces calibrated benchmark scores that answer the central evaluation question: *what would this model's accuracy be had it never seen the test data?* On the psychometric side, item response theory (IRT) has been increasingly applied to LLM evaluation. PSN-IRT [Gu et al., 2026] models item difficulty and discrimination but does not account for contamination. Polo et al. [2026] validate IRT at the scale of 400+ models yet treat all items as equally trustworthy. ATLAS [Lyu et al., 2025] and Lalor et al. [2024] further develop IRT-based evaluation without contamination modeling, and TinyBenchmarks [Polo et al., 2024] demonstrates that carefully selected item subsets can approximate full benchmark scores, but again without contamination awareness. HiBayES [Zhuang et al., 2025] provides hierarchical Bayesian uncertainty quantification but ignores item compromise. Inference-time decontamination [Yang et al., 2024] and targeted probing [Han et al., 2026] attempt to adjust or detect contamination effects at evaluation time, but neither produces fully calibrated scores with uncertainty quantification. The closest predecessor, DCR [Li et al., 2025], provides adjusted accuracy through fuzzy inference over detection signals but offers no uncertainty bounds and does not address the systematic signal degradation under RL post-training that we document in this work.

The case for post-hoc calibration rests on a sobering empirical finding: contamination prevention does not work. Xu et al. [2025b] systematically evaluate twenty mitigation strategies—including item rephrasing, option shuffling, filtering, and dynamic benchmark construction—and find that none reliably eliminates contamination effects across models and benchmarks. Zhang et al. [2025] further demonstrate that standard decontamination procedures fail to fully remove memorized content from model weights. While dynamic benchmarks such as LiveBench [White et al., 2024] sidestep contamination by continuously generating new items, and efficient evaluation via item subsampling [Polo

et al., 2024] can mitigate item-specific biases, neither approach provides calibrated scores for existing contaminated benchmarks. These findings motivate a paradigm shift from contamination *elimination* to contamination *adjustment*: rather than attempting to remove a confounder that resists removal, we measure it and statistically control for it. The analogy to observational causal inference is instructive—when randomization is infeasible, valid estimates are obtained not by eliminating confounders but by measuring and adjusting for them under clearly stated assumptions. ContamCal instantiates this "statistical safety net" philosophy for benchmark evaluation.

Our approach is grounded in educational measurement's longstanding treatment of item compromise—the leaking of test content to examinees prior to administration. Item compromise IRT models [Eckerly, 2017, Sinharay, 2017] incorporate exposure indicators as covariates within the item response function, enabling joint estimation of compromise effects alongside standard item and person parameters. Explanatory IRT [De Boeck and Wilson, 2004, Adams et al., 1997] provides the broader theoretical framework for embedding external covariates in item response models. Adapting this psychometric tradition to LLM evaluation requires three non-trivial structural modifications. First, whereas educational IRT assumes independent examinees, LLM evaluation involves individual models producing all responses to a shared item pool, fundamentally altering local independence semantics [Mislevy, 1986]. Second, binary leakage indicators (compromised vs. uncompromised) must be replaced with continuous exposure scores derived from noisy, imperfect detection signals—necessitating a measurement error model that propagates detection uncertainty into ability estimates. Third, stable examinee ability must give way to regime-conditional ability, as models undergoing supervised fine-tuning (SFT) versus RL post-training exhibit qualitatively different contamination signatures that affect both detection signal reliability and contamination effect magnitude.

We propose ContamCal, a two-stage post-hoc calibration framework for contamination-adjusted benchmark scoring. In Stage 1 (*Multi-Signal Provenance Mapping*), four training-free detection signals—embedding-based near-duplicate retrieval, Min-K%++ membership inference, ConStat performance gaps, and Self-Critique entropy collapse—are aggregated via reciprocal rank fusion [RRF; Cormack et al., 2009] to produce per-item exposure scores. RRF is a deliberate design choice: a learned aggregator would face meta-circularity, as its own training data may be contaminated, precluding unbiased weight estimation. In Stage 2 (*Contamination-Augmented 4PL IRT*), exposure scores enter a four-parameter logistic IRT model as covariates, with a global contamination effect parameter γ estimated jointly with item and ability parameters via Bayesian inference. A measurement error model treats the observed exposure scores as noisy proxies for latent contamination status, automatically propagating detection uncertainty into posterior estimates. Calibrated scores are obtained by setting exposure to zero and marginalizing over parameter uncertainty, producing point estimates with 50%/80%/95% Bayesian credible intervals. Models trained via RL receive appropriately wider intervals, reflecting the genuine epistemic uncertainty that arises when detection signals are degraded.

Our contributions are organized hierarchically. At the core: (1) the ContamCal framework itself—a two-stage pipeline comprising RRF-based provenance mapping and contamination-augmented 4PL IRT with per-benchmark contamination effects and a measurement error model for exposure—that produces calibrated benchmark scores with Bayesian credible intervals; (2) a signal survival analysis demonstrating that contamination detection signals degrade in a cliff-like, rather than gradual, manner under RL post-training, yielding a signal $\times$ regime AUC mapping with practical implications for the detection community; and (3) dual-track validation through controlled SFT contamination injection across multiple model scales (0.5B–7B) and families (internal validity) combined with an ecological RL case study (external validity). Supporting these core contributions: (4) a simulation identifiability study grounded in Fisher information analysis, with a 144-condition sweep establishing minimum signal quality and model count thresholds for reliable parameter recovery; and (5) comprehensive evaluation against seven baselines using five quantitative metrics—calibration MAE, credible interval coverage, interval sharpness, ranking preservation (Kendall's τ), and dose-response monotonicity—with bootstrap confidence intervals on all comparisons.

2 Related Work

TODO: Related work to be written.

3 Method

ContamCal operates in two stages. Stage 1 (§3.1) estimates per-item contamination exposure by fusing multiple detection signals into a single score. Stage 2 (§3.2–§3.4) embeds these scores as covariates in a contamination-augmented IRT model, performing Bayesian inference to produce calibrated benchmark scores with credible intervals.

3.1 Stage 1: Multi-Signal Provenance Mapping

For each model i and benchmark item j, we compute four complementary detection signals that capture different facets of contamination evidence.

Embedding-based near-duplicate retrieval. We embed both training corpus segments and benchmark items using a sentence transformer encoder and build a FAISS index over training embeddings for efficient retrieval. The signal $s_1(i, j)$ is the maximum cosine similarity between item j and the top-K retrieved training segments for model i's training corpus, capturing semantic overlap that survives paraphrasing and surface-level reformulation.

Min-K%++ membership inference. Following Duan et al. [2025], we compute the average log-probability of the $k\%$ least-likely tokens in each benchmark item under model i. Contaminated items yield systematically higher (less negative) Min-K%++ scores, as the model assigns elevated probability to tokens it has memorized during training. The signal $s_2(i, j)$ is the resulting membership inference statistic.

ConStat performance gap. For each benchmark item, we generate a semantically equivalent paraphrased variant using an independent rewriter model. The signal $s_3(i, j)$ is the accuracy gap between model i's performance on the original item versus its paraphrased counterpart [De Koninck et al., 2024]. A model that has memorized the original surface form but genuinely understands the underlying concept should perform comparably on both versions; a large gap indicates surface-level memorization rather than robust capability.

Self-Critique entropy collapse. When model i is prompted to evaluate the correctness of its own response to item j, contaminated items elicit low-entropy self-evaluation distributions—the model exhibits overconfidence in its assessment of memorized content [Wüst et al., 2025]. The signal $s_4(i, j)$ is the negative entropy of the self-critique distribution. This signal is particularly valuable because it operates at the meta-cognitive level and exhibits greater robustness to RL post-training than signals relying on token-level statistics (§3.1).

Signal survival analysis. Detection signals do not degrade uniformly across training regimes. Under SFT post-training, all four signals retain moderate to high discriminative power (AUC > 0.7 in controlled experiments). Under RL post-training with GRPO [Yang et al., 2026], however, signals s_1–s_3 exhibit cliff-like degradation—dropping to near chance-level AUC—while the Self-Critique signal (s_4) retains substantially greater discriminative power. This signal $\times$ regime interaction has two practical consequences. First, exposure estimates for RL-trained models carry higher uncertainty, which the measurement error model in Stage 2 propagates into appropriately wider credible intervals. Second, the cliff-like (rather than gradual) degradation pattern implies that RL does not merely attenuate detection signals but fundamentally restructures the model's relationship to memorized content. We report the full signal survival AUC matrix in §**??**.

RRF aggregation. We aggregate the four signals via reciprocal rank fusion [Cormack et al., 2009]. For each model i, items are ranked independently by each signal s, and the fused exposure score is:

$$e_{ij} = \sum_{s=1}^{4} \frac{1}{k + r_{s,i}(j)}, \tag{1}$$

where $r_{s,i}(j)$ is the rank of item j under signal s for model i, and $k = 60$ is the standard RRF smoothing constant. RRF is a deliberate design choice over learned aggregation (*e.g.*, logistic regression over signal scores). A learned aggregator requires labeled training data indicating which items are truly contaminated—but ground-truth contamination labels are precisely what we lack, and the aggregator's training data may itself be contaminated, creating meta-circularity. RRF avoids this by requiring no training, providing a robust, parameter-free fusion baseline.

We note that KDS [Xu et al., 2025a] provides a complementary detection signal based on knowledge distillation signatures, but requires per-model LoRA fine-tuning at $O(N^2)$ computational cost. We include KDS in ablation experiments but exclude it from the default pipeline for scalability.

3.2 Stage 2: Contamination-Augmented 4PL IRT

Model specification. We augment the standard four-parameter logistic (4PL) IRT model [Lord, 1980] with a contamination covariate. The probability that model i correctly answers item j is:

$$P(y_{ij} = 1 \mid \boldsymbol{\psi}) = c_j + (d_j - c_j)\,\sigma\big(a_j(\theta_i - b_j) + \gamma \cdot \tau_{ij}\big), \tag{2}$$

where $\sigma(\cdot)$ denotes the logistic function, $y_{ij} \in \{0, 1\}$ is the binary response, and $\boldsymbol{\psi}$ collects all parameters. The standard IRT parameters are: $\theta_i \in \mathbb{R}$ (ability of model i), $a_j > 0$ (discrimination of item j), $b_j \in \mathbb{R}$ (difficulty of item j), $c_j \in [0, 1]$ (guessing/lower asymptote), and $d_j \in [0, 1]$ (upper asymptote/slip parameter). The contamination-specific parameters are: $\gamma \in \mathbb{R}$ (global contamination effect) and $\tau_{ij} \geq 0$ (latent true exposure of model i to item j).

The additive structure $a_j(\theta_i - b_j) + \gamma \cdot \tau_{ij}$ admits a natural interpretation: contamination exposure shifts the effective ability of a model on a given item by $\gamma \cdot \tau_{ij}/a_j$ units on the ability scale. When $\tau_{ij} = 0$ (no contamination), Eq. (2) reduces to the standard 4PL model. A positive γ indicates that contamination inflates performance, as expected.

Measurement error model. The latent exposure τ_{ij} is not directly observed; we observe the noisy RRF-derived estimate e_{ij} from Stage 1. We model this relationship as:

$$\tau_{ij} \sim \mathcal{N}(e_{ij},\, \sigma_e^2), \tag{3}$$

where σ_e^2 is the measurement error variance, estimated jointly with other model parameters. This formulation treats the observed exposure scores as noisy proxies for latent contamination and has two critical consequences. First, uncertainty in the detection signals is automatically propagated into the posterior over ability θ_i and contamination effect γ, yielding appropriately wider credible intervals when signals are unreliable. Second, under RL post-training—where signal degradation is severe (§3.1)—the posterior distribution of τ_{ij} becomes more diffuse, producing wider credible intervals that honestly reflect our reduced ability to distinguish contaminated from clean items.

Prior specification. We assign the following priors:

$$\theta_i \sim \mathcal{N}(0, 1), \tag{4}$$
$$a_j \sim \text{LogNormal}(0, 0.5), \tag{5}$$
$$b_j \sim \mathcal{N}(0, 1), \tag{6}$$
$$c_j \sim \text{Beta}(5, 20), \tag{7}$$
$$d_j \sim \text{Beta}(20, 5), \tag{8}$$
$$\gamma \sim \mathcal{N}(0, 1). \tag{9}$$

The $\text{Beta}(5, 20)$ prior on c_j encodes the expectation that guessing rates are low (mean ≈ 0.2) but non-negligible for multiple-choice items. The $\text{Beta}(20, 5)$ prior on d_j reflects the expectation that upper asymptotes are high (mean ≈ 0.8) but not necessarily unity, allowing for careless errors or ambiguous items. The $\text{LogNormal}(0, 0.5)$ prior on a_j constrains discrimination to be positive with moderate magnitude (median $= 1$). The weakly informative $\mathcal{N}(0, 1)$ prior on γ permits both positive and negative contamination effects; we conduct sensitivity analyses under $\mathcal{N}(0, 0.5)$ and $\mathcal{N}(0, 2)$ to assess robustness to this specification (§**??**).

Connection to item compromise IRT. Equation (2) extends the item compromise IRT tradition [Eckerly, 2017, Sinharay, 2017] with three structural adaptations for the LLM evaluation setting.

Adaptation 1: Independent examinees → shared item pool. In educational testing, each examinee independently samples from a population ability distribution. In LLM evaluation, a single model responds to all items, and its responses are correlated through shared model parameters. This alters the local independence assumption: conditional on θ_i, item responses within a single model may remain correlated through model-specific idiosyncrasies not captured by the scalar ability parameter [Mislevy, 1986, San Martín et al., 2009]. We mitigate this by fitting IRT models per-benchmark—reducing the scope of potential violations—and empirically assessing residual correlations via posterior predictive checks (§**??**).

Adaptation 2: Binary leakage → continuous exposure. Classical item compromise models use binary indicators $e_{ij} \in \{0, 1\}$ [Sinharay, 2017]. We generalize to continuous exposure scores $\tau_{ij} \in \mathbb{R}^+$ derived from noisy detection

signals, which is standard in explanatory IRT [De Boeck and Wilson, 2004] but necessitates the measurement error formulation of Eq. (3) to prevent attenuation bias from regression dilution [Maris and Bechger, 2009].

Adaptation 3: Stable ability → regime-conditional ability. An examinee's ability is typically assumed stable across test items. For LLMs, the training regime (SFT vs. RL) qualitatively changes both the model's ability profile and the reliability of detection signals. We handle this through the signal survival analysis (§3.1), which informs the measurement error variance, rather than through regime-specific ability parameters—preserving model parsimony while acknowledging regime effects through the uncertainty channel.

3.3 Bayesian Inference

We perform full Bayesian inference using the No-U-Turn Sampler [NUTS; Hoffman and Gelman, 2014] implemented in CmdStanPy [Carpenter et al., 2017]. We run four independent Markov chains and assess convergence via three diagnostics: split-$\hat{R} < 1.01$ for all parameters, effective sample size ESS > 400 per parameter, and fewer than 0.1% divergent transitions. The target acceptance probability is set to `adapt_delta` $= 0.95$ to reduce divergences in the high-curvature posterior geometry induced by the 4PL model's bounded response function.

Per-benchmark fitting. Rather than fitting a single IRT model jointly across all benchmarks, we fit separate models per benchmark. This design choice reduces the scope of local independence violations: within a single benchmark, items share a common knowledge domain, making the conditional independence assumption more defensible than across heterogeneous benchmarks encompassing knowledge recall, mathematical reasoning, and code generation [San Martín et al., 2009]. Per-benchmark fitting also permits the benchmark-specific contamination effect variant γ_k described in §3.4.

Few-examinee regime. Standard IRT applications assume large examinee populations ($N > 200$), but current LLM evaluation studies may include as few as 20–50 models. In this low-N regime, item parameter estimation becomes unstable without informative priors. We address this by incorporating empirical Bayes priors for item parameters derived from the Growing Pains study [Polo et al., 2026], which provides item parameter estimates from a pool of 400+ models. Specifically, the item difficulty and discrimination priors are centered at the Growing Pains point estimates with moderately inflated variances, serving as a regularizing anchor while permitting data-driven updating. We validate this prior transfer procedure via simulation-based calibration [SBC; Talts et al., 2018], confirming that posteriors are well-calibrated under the specified priors across a range of sample sizes (§**??**). Automatic differentiation variational inference [ADVI; Ulitzsch and Nestler, 2022] is explored as a scalability option for larger item pools in the appendix.

3.4 Calibrated Score Computation

Given the posterior distribution over all parameters, we compute the calibrated (decontaminated) accuracy for model i on benchmark B by setting exposure to zero and marginalizing over posterior uncertainty:

$$\text{Acc}_i^{\text{cal}} = \frac{1}{|B|} \sum_{j \in B} \mathbb{E}_{\boldsymbol{\psi} \sim p(\boldsymbol{\psi}|\mathbf{y})} \left[c_j + (d_j - c_j)\, \sigma\big(a_j(\theta_i - b_j)\big) \right], \tag{10}$$

where the expectation is taken over posterior draws of $\boldsymbol{\psi}$ and the contamination term $\gamma \cdot \tau_{ij}$ has been removed. This counterfactual quantity answers: *what would model i's expected accuracy be on benchmark B had it never been exposed to any test items during training?*

In practice, we approximate Eq. (10) by averaging over L posterior draws $\{\boldsymbol{\psi}^{(\ell)}\}_{\ell=1}^{L}$:

$$\widehat{\text{Acc}}_i^{\text{cal}} = \frac{1}{L} \sum_{\ell=1}^{L} \frac{1}{|B|} \sum_{j \in B} \left[c_j^{(\ell)} + (d_j^{(\ell)} - c_j^{(\ell)})\, \sigma\big(a_j^{(\ell)}(\theta_i^{(\ell)} - b_j^{(\ell)})\big) \right]. \tag{11}$$

Credible intervals at level α are obtained from the empirical quantiles of $\{\text{Acc}_i^{\text{cal},(\ell)}\}_{\ell=1}^{L}$. We report 50%, 80%, and 95% intervals. By construction, models with higher exposure scores and noisier detection signals—whose τ_{ij} posteriors are more diffuse—receive wider credible intervals, encoding the principle that calibration is only as reliable as its inputs.

Per-benchmark contamination effects. The primary model assumes a global γ shared across all benchmarks. As an ablation, we consider a per-benchmark variant:

$$P(y_{ij} = 1 \mid \psi_k) = c_j + (d_j - c_j)\,\sigma\big(a_j(\theta_i - b_j) + \gamma_k \cdot \tau_{ij}\big), \tag{12}$$

where γ_k is the contamination effect specific to benchmark k, with independent priors $\gamma_k \sim \mathcal{N}(0, 1)$. This variant can capture heterogeneous contamination effects—for instance, knowledge-intensive benchmarks (MMLU) may exhibit larger contamination inflation than multi-step reasoning benchmarks (GSM8K)—at the cost of reduced statistical power per benchmark. We compare both specifications in §**??**.

References

Raymond J. Adams, Mark Wilson, and Wen-chung Wang. The multidimensional random coefficients multinomial logit model. *Applied Psychological Measurement*, 21(1):1–23, 1997.

Bob Carpenter, Andrew Gelman, Matthew D. Hoffman, Daniel Lee, Ben Goodrich, Michael Betancourt, Marcus Brubaker, Jiqiang Guo, Peter Li, and Allen Riddell. Stan: A probabilistic programming language. *Journal of Statistical Software*, 76(1):1–32, 2017.

Yupeng Chang, Xu Wang, Jindong Wang, Yuan Wu, Linyi Yang, Kaijie Zhu, Hao Chen, Xiaoyuan Yi, Cunxiang Wang, Yidong Wang, Wei Ye, Yue Zhang, Yi Chang, Philip S. Yu, Qiang Yang, and Xing Xie. A survey on evaluation of large language models. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2025.

Gordon V. Cormack, Charles L. A. Clarke, and Stefan Buettcher. Reciprocal rank fusion outperforms Condorcet and individual rank learning methods. In *Proceedings of the 32nd International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 758–759. ACM, 2009.

Paul De Boeck and Mark Wilson. *Explanatory Item Response Models: A Generalized Linear and Nonlinear Approach*. Springer, New York, 2004.

Jasper De Koninck, Koen Nelissen, Siegfried Nijssen, and Jesse Davis. ConStat: Performance-based contamination detection in large language models. In *Advances in Neural Information Processing Systems 37 (NeurIPS)*, 2024.

Yifan Dong, Xinran Li, Difan Chen, and Zhengyu Wang. Dose-response relationships in benchmark contamination: Quantifying score inflation across model scales. *arXiv preprint*, 2026.

Jiawen Duan, Cheng Xie, Jian Zhang, and Kai Li. Min-K%++: Improved membership inference for detecting pre-training data of large language models. In *Proceedings of the 13th International Conference on Learning Representations (ICLR)*, 2025.

Carol Eckerly. Detecting item compromise in the context of item response theory. *Applied Psychological Measurement*, 41(2):87–101, 2017.

Jiaqi Gu, Hainiu Wang, Qingyao Ye, and Nanyun Peng. PSN-IRT: Psychometric-informed evaluation of large language models via item response theory. In *Proceedings of the 40th AAAI Conference on Artificial Intelligence*, 2026.

Yucheng Han, Chongxuan Wu, Jiashi Qi, and Jie Zhou. Targeted probing for contamination effects in large language model evaluations. In *Findings of the 2026 Conference of the European Chapter of the Association for Computational Linguistics (EACL Findings)*, 2026. arXiv:2510.07175.

Matthew D. Hoffman and Andrew Gelman. The No-U-Turn sampler: Adaptively setting path lengths in Hamiltonian Monte Carlo. *Journal of Machine Learning Research*, 15(47):1593–1623, 2014.

John P. Lalor, Yi Yang, and Noah A. Smith. Benchmarking large language models with item response theory. In *Proceedings of the 38th AAAI Conference on Artificial Intelligence*, 2024.

Yucheng Li, Frank Flanigan, Tom Sherborne, and Charlie Sherborne. DCR: Decontaminated calibration and ranking for benchmark evaluation. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2025.

Frederic M. Lord. *Applications of Item Response Theory to Practical Testing Problems*. Lawrence Erlbaum Associates, Hillsdale, NJ, 1980.

Haoran Lyu, Yufei Ding, Hainiu Wang, and Nanyun Peng. ATLAS: Adaptive testing via latent ability scoring for large language models. *arXiv preprint*, 2025.

Gunter Maris and Timo Bechger. On interpreting the model parameters for the three parameter logistic model. *Measurement*, 7(2):75–88, 2009.

Robert J. Mislevy. Bayes modal estimation in item response models. *Psychometrika*, 51(2):177–195, 1986.

Yonatan Oren, Nicole Meister, Niladri Chatterji, Faisal Laesecke, and Ryan Cotterell. Proving test set contamination in black box language models. In *Proceedings of the 12th International Conference on Learning Representations (ICLR)*, 2024.

Felipe Maia Polo, Lucas Weber, Leshem Choshen, Yuekai Sun, Gongjun Xu, and Mikhail Yurochkin. TinyBenchmarks: Evaluating LLMs with fewer examples. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, 2024.

Felipe Maia Polo, Ronald Mamani, Yuekai Zhuang, Hao Xu, Leshem Choshen, Mikhail Yurochkin, and Yuekai Sun. Growing pains: Understanding the limitations and potential of IRT for LLM evaluation at scale. *arXiv preprint*, 2026.

Sumedh Rao, Inioluwa Deborah Raji, and Percy Liang. The silicon bureaucracy: Evaluating the evaluators in LLM benchmarking. *arXiv preprint*, 2026.

Ernesto San Martín, Guido Del Pino, and Paul De Boeck. Consequential validity of item response theory models for measuring latent constructs. *Psychometrika*, 74(2):349–367, 2009.

Sandip Sinharay. Detection of item preknowledge using likelihood ratio test and score test. *Journal of Educational and Behavioral Statistics*, 42(1):46–68, 2017.

Sean Talts, Michael Betancourt, Daniel Simpson, Aki Vehtari, and Andrew Gelman. Validating Bayesian inference algorithms with simulation-based calibration. *arXiv preprint*, 2018.

Esther Ulitzsch and Steffen Nestler. Automatic differentiation variational inference for item response theory models. *British Journal of Mathematical and Statistical Psychology*, 75(3):421–447, 2022.

Yuxin Wang, Yucheng Zhao, Cheng Li, and Meng Zhang. On the fragility of benchmark evaluations for large language models. *arXiv preprint*, 2025.

Colin White, Samuel Dooley, Manley Roberts, Arka Pal, Benjamin Feuer, Siddhartha Jain, Ravid Shwartz-Ziv, Neel Joshi, and Micah Goldblum. LiveBench: A challenging, contamination-free LLM benchmark. *arXiv preprint*, 2024.

Timon Wüst, Jan Hoffmann, and Richard Yuanzhe Pang. Self-critique as a contamination signal: Detecting memorization via entropy collapse in model self-evaluations. *arXiv preprint*, 2025.

Yanyang Xu, Hongzhan Qi, Benyou Zhao, and Zhiyuan Liu. Knowledge distillation signatures for efficient contamination detection. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, 2025a. arXiv:2502.00678.

Yiwen Xu, Kaijie Song, Jianfeng Zhou, and Liang Chen. The emperor's new clothes: Systematic evaluation of contamination mitigation strategies for large language models. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, 2025b. arXiv:2503.16402.

Shuo Yang, Xin Wei, and Weicheng Shi. Inference-time decontamination of large language models for benchmark evaluation. In *Findings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP Findings)*, 2024.

Shuo Yang, Xin Wei, Zhiwei Zhang, and Feiyu Xu. Reinforcement learning amplifies contamination in LLM benchmark evaluation. *arXiv preprint*, 2026.

Yiming Zhang, Yuxin Wang, Hao Dong, and Jiashi Li. How much can we forget? examining decontamination effectiveness in large language models. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, 2025.

Yuekai Zhuang, Felipe Maia Polo, Mikhail Yurochkin, and Yuekai Sun. HiBayES: Hierarchical Bayesian evaluation system for language models. *arXiv preprint*, 2025.